Systemic protein-network dependencies predict chronic kidney disease

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Chronic kidney disease (CKD) often develops without early symptoms, making prospective risk stratification important [1]. The central scientific question is whether CKD-related molecular risk is captured by individual protein concentrations or also by the patient-specific dependencies linking metabolic, vascular and inflammatory systems. We therefore test whether the network-dependent component of a protein measurement contains information about future CKD that its abundance and prior diagnoses do not capture. This distinction motivates a directed-network representation of systemic molecular organization [2].

Baseline proteomics supplies a single snapshot per participant, with no observed molecular time course. To model how protein dependencies vary across proteomic states, we introduce an H-index derived from aggregate protein abundance. This common abundance coordinate converts cross-sectional variation into response curves, making quasi-dynamic modelling possible without repeated protein measurements. Under idopNetwork assumptions [3, 4], these curves are decomposed into intrinsic and dependency-associated components to estimate participant-specific network effects. The aim is to represent state-dependent molecular relationships not explicitly represented by individual protein abundances or average correlations. H-index is a modelling coordinate, not elapsed time or an established disease-progression axis.

We analyse 39,047 UK Biobank participants with longitudinal electronic health records (EHRs) and 2,919 plasma proteins. Functional clustering [5] defines 32 modules. A hierarchical Graph Transformer integrates directed dependencies within and between modules (Fig. 1) with clinical trajectories encoded by a frozen Delphi-based Transformer. Prediction-guided gating selects a 60-protein induced subgraph for retraining.

In 7,794 formal-validation participants, the Core Graph achieves six-year areas under the receiver operating characteristic (AUROC) and precision–recall (AUPR) curves of 0.820 and 0.206, compared with 0.751 and 0.107 for diagnosis history (Fig. 3). The selected proteins span 20 modules. With EHR inputs fixed, retaining only edge-information channels yields AUROC/AUPR of 0.818/0.184, versus approximately 0.774/0.140 for abundance alone. These channels combine individual edges, total incoming effects and inter-module effects. Reassigning edge profiles between participants reduces discrimination towards diagnosis-only performance. EHR-derived phenotypes distinguish CKD incidence (Fig. 2).

These results support the predictive relevance of individualized protein dependencies. Quasi-dynamic representation makes a baseline proteomic snapshot usable for estimating state-dependent network features, without establishing temporal or causal mechanisms. The compact protein set is a candidate for targeted panels and risk-adapted renal surveillance, requiring external validation and prospective evaluation before clinical use.

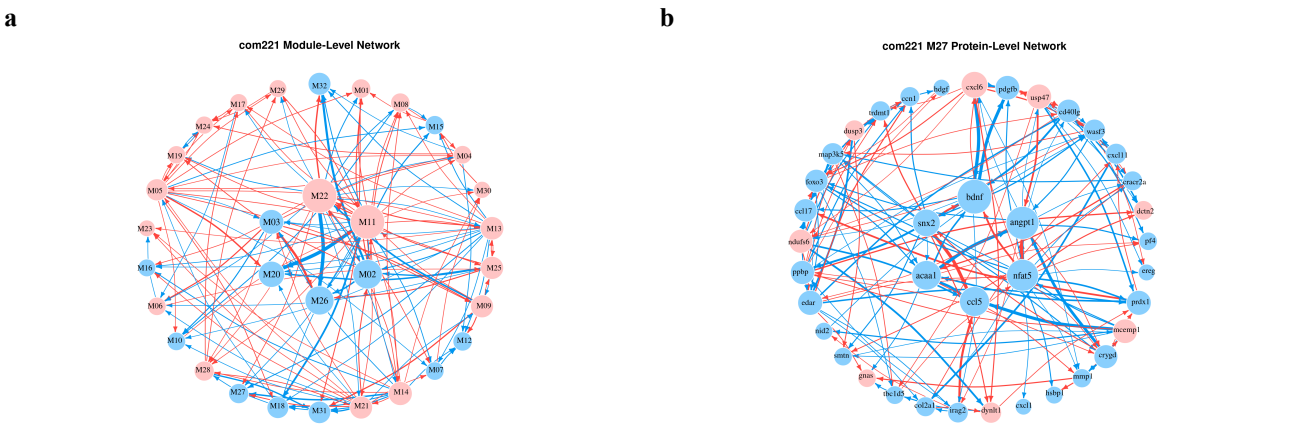


Figure 1: Directed protein networks at module (a) and within-module (b) scales.

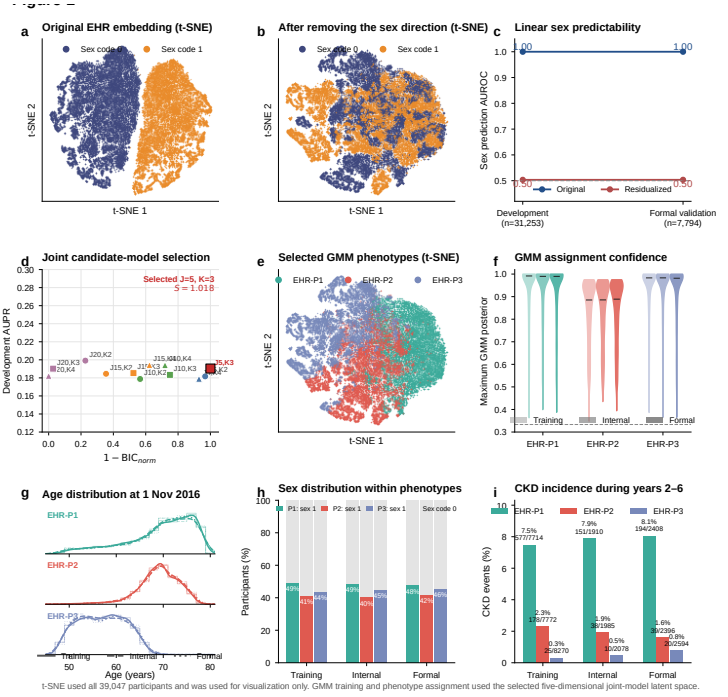


Figure 2: EHR phenotyping after removal of the leading linear sex direction: embeddings, model selection, assignment confidence and clinical characteristics.

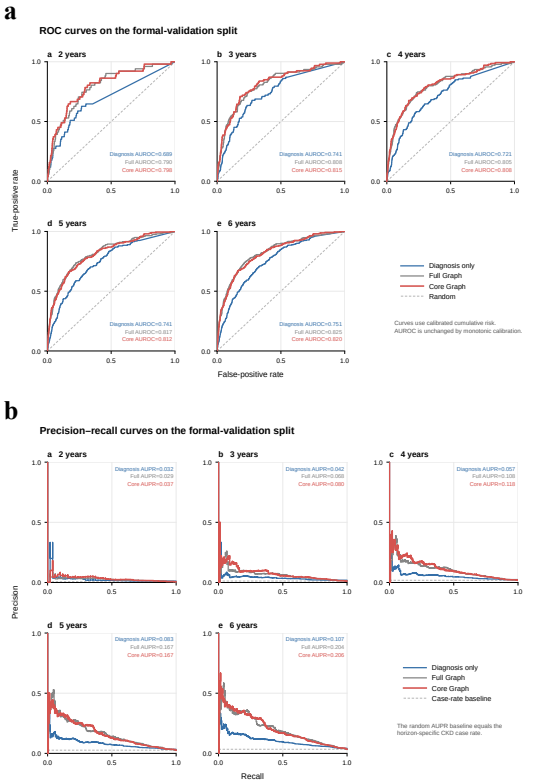


Figure 3: ROC (a) and precision–recall (b) curves on the formal-validation set.

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